

ROHAN KUMAR

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PROFESSIONAL SUMMARY

Senior Data Engineer & Analytics Specialist with over 16 years of quantitative systems experience designing and operating scalable cloud and on-premises data architecture, production ETL/ELT pipelines, and relational/analytical data models that turn raw data into actionable insights. Proven track record architecting end-to-end pipelines — from raw source through transformation, validation, and production handoff — for market data, transactions, and time-series datasets across global platforms including SMBC, Deutsche Bank, HSBC, and Santander. Databricks Certified Data Engineer skilled in Python, SQL, PySpark, Azure/AWS, and data modeling for semantic/analytical layers, with a consistent focus on tested, versioned, observable code over one-off scripts. Comfortable working with ambiguity and shifting requirements; adept at translating business needs into technical specifications while partnering with Quantitative Researchers, Portfolio Managers, Traders, and cross-functional Tech teams.

CORE COMPETENCIES

- **Data Engineering & Architecture:** Cloud & On-Prem Data Architecture Design, Production ETL/ELT Pipelines, Data Modeling (Relational, Analytical & Semantic Layer), Data Integration, Data Security, Data Warehousing, Data Validation & Quality Monitoring
- **Engineering Practices:** Tested & Versioned Production Code, CI/CD, Schema/Data Validation, Bias & Leakage Checks, Complex Problem-Solving, Cross-Functional Stakeholder Collaboration
- **Programming & Tools:** Python (OOP, Pandas, NumPy), SQL, PostgreSQL, JavaScript, PySpark, kdb+/q, Docker, Git/GitHub, JIRA
- **Cloud & Big Data:** Azure Databricks, AWS, Snowflake, Kafka, Redis, EKS/Kubernetes, dbt
- **Domain Expertise:** Market Data Ingestion, Fixed Income & FX Transactions, Time-Series Modeling, Quantitative Research Support, Front-Office Trading Systems

PROFESSIONAL EXPERIENCE

Madison-Davis / SMBC Compliance, Jersey City, NJ

Senior Data Engineer – Investment & Surveillance Analytics | Jun 2025 – Dec 2025

- Designed and maintained automated production ETL pipelines in Azure Databricks (Python/SQL) ingesting high-volume Fixed Income and FX tick and transaction data, cutting daily manual processing time by 20–30%.
- Architected relational, analytical, and semantic-layer data models in PostgreSQL/Databricks — owning the pipeline from raw source through transformation to production handoff — enabling sub-second surveillance analytics and multi-asset trade pattern detection.
- Implemented automated data validation, anomaly and bias detection, sample-size checks, and data reconciliation to monitor data quality and confirm metrics answered the intended question across incoming operational feeds and market data streams.
- Partnered with Trading Desk SMEs, Quantitative Specialists, and Tech teams to ingest, enrich, and transform large-scale datasets while maintaining full data lineage and auditability.
- Enforced software engineering best practices — writing tested, versioned, and observable production code rather than one-off notebooks — including unit testing, Git version control, continuous integration, and technical runbook documentation in Confluence.

PHD / M.S. Computer Science – Applied Engineering Research, Jersey City, NJ

Lead Data Engineer & Quantitative Systems Researcher | Jan 2024 – Present

- Architected an automated real-time market data ingestion engine in Python (OOP), streaming and processing tick and order book datasets via financial APIs into a normalized PostgreSQL schema.

- Developed browser-based internal reporting interfaces and dashboards using JavaScript and Databricks SQL, streamlining metrics visibility for operations and quantitative analysis.
- Optimized analytical SQL queries and transformations for high-frequency time-series datasets, reducing end-to-end analytical latency.
- Published applied ML and anomaly detection research on SSRN (Abstract ID 6576018), evaluating high-throughput transactional data integrity and pattern discovery across asset classes.

Deutsche Bank, New York, NY

AVP – Quantitative Data & Risk Platform Engineer | Aug 2022 – Dec 2023

- Engineered scalable data pipelines supporting enterprise forecasting engines and IRRBB analytics, aggregating over \$5B in revenue and balance sheet exposures.
- Maintained complex relational database structures and automated SQL transformation logic across Institutional Cash Management and Corporate Trust datasets.
- Integrated automated testing, schema validation, and reconciliation scripts that reduced model reporting turnaround and review cycles by 30%.
- Analyzed business requirements from portfolio modeling teams, front-office traders, and enterprise data architects, translating them into technical specifications and data architecture decisions that ensured pipeline reliability and data availability under rigorous regulatory environments.

Primesoft Consulting Services (Santander), Jersey City, NJ

Data & Analytics Engineer (Contract) | Apr 2020 – Aug 2022

- Built robust data extraction, transformation, and load (ETL) pipelines using SQL and SAS, processing millions of retail loan and transactional records for credit modeling engines.
- Designed analytical data marts and automated data validation workflows, improving data accuracy across enterprise risk reporting by 98%.
- Created structured datasets and managed schema updates across staging and analytical database environments.

Huxley / Allegis Global Solution (HSBC), New York, NY

Analytics & Data Platform Specialist (Contract) | Jun 2018 – Nov 2019

- Automated data feed validation and pipeline health monitoring, maintaining 99% operational compliance and data integrity across transaction monitoring feeds.
- Partnered with IT engineering teams to implement production data validation scripts, UAT environments, and comprehensive technical specifications.

CRISIL / S&P Global, New York, NY

Quantitative Data Analyst | Sep 2015 – Nov 2017

- Designed time-series ETL routines and staging tables for Tier-1 investment banks (Citi, BofA, BNP), cutting analytical data preparation latency by 20%.
- Validated and cataloged complex financial market datasets across macroeconomic, credit, and equity indicators.

Kodukulla and Associates Inc., Chicago, IL

Quantitative Research & Data Analyst | Jan 2015 – May 2015

- Developed quantitative simulation data models using Monte Carlo methods to evaluate cash flow scenarios and project valuation distributions.

EARLIER CAREER – TECHNOLOGY & ENGINEERING

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- HCL Technologies – Lead Software Engineer (May 2012 – Jul 2013)
 - Tata Motors – Systems / Quality Engineer (Jan 2009 – Nov 2011)
 - Tech Mahindra – Software Developer (Dec 2007 – Dec 2008)
 - Action Construction Equipment – Design Engineer (Sep 2005 – Dec 2007)

EDUCATION

- Ph.D., Computational Science and Engineering (In Progress, Expected 2029) | Harrisburg University of Science and Technology, PA
- M.S., Computer Science (GPA: 4.0) (2025-2026) | Monroe University, New Rochelle, NY
- M.S., Analytics (GPA: 3.9) (2018-2019) | Harrisburg University of Science and Technology, PA
- M.S., Finance – Quantitative Risk & Corporate Finance (2013-2015) | Illinois Institute of Technology, Chicago, IL
- B.E., Mechanical Engineering (2001-2005) | Manipal Institute of Technology, India

CERTIFICATIONS & WORK AUTHORIZATION

- **Certifications:** Databricks Certified Data Engineer Associate | Equity Essentials, Bloomberg Training Program
- **Work Authorization:** F-1 PhD Student in Computational Science & Engineering. Eligible for immediate CPT work authorization requiring zero employer sponsorship.